

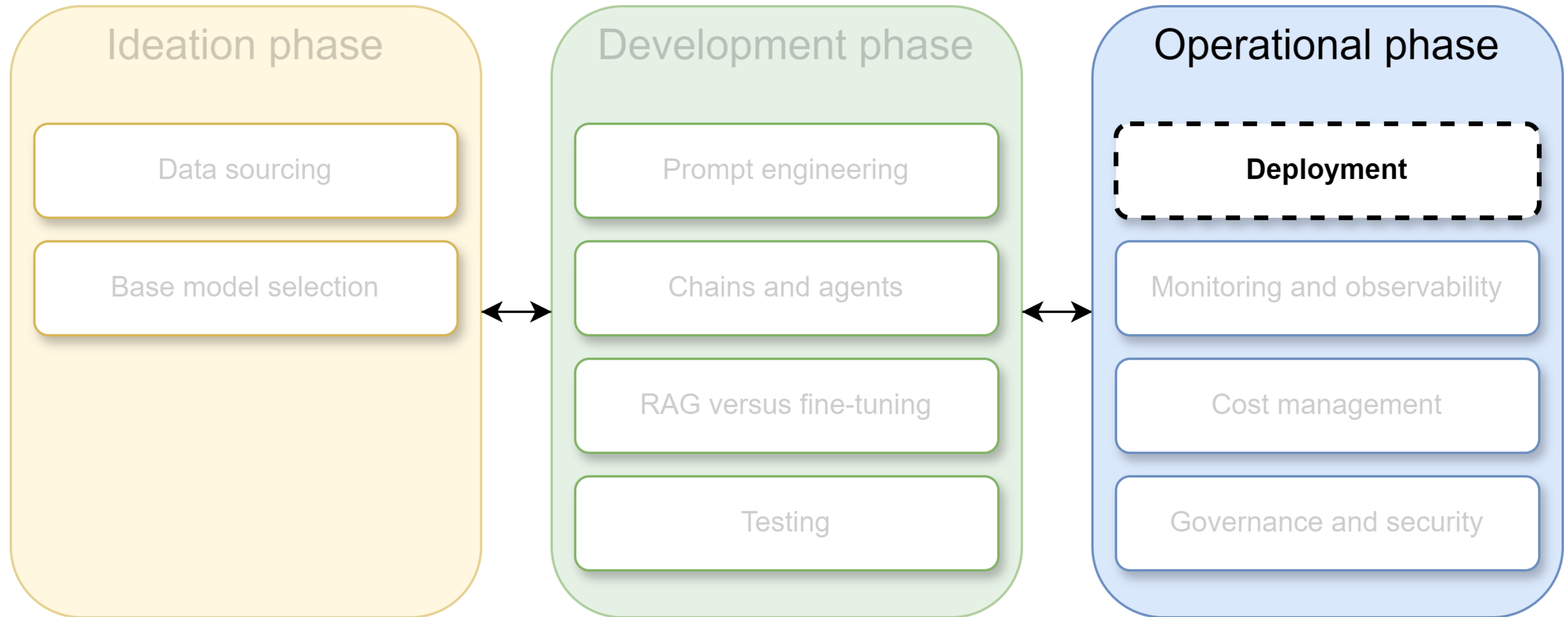
Deployment

LLMOPS CONCEPTS

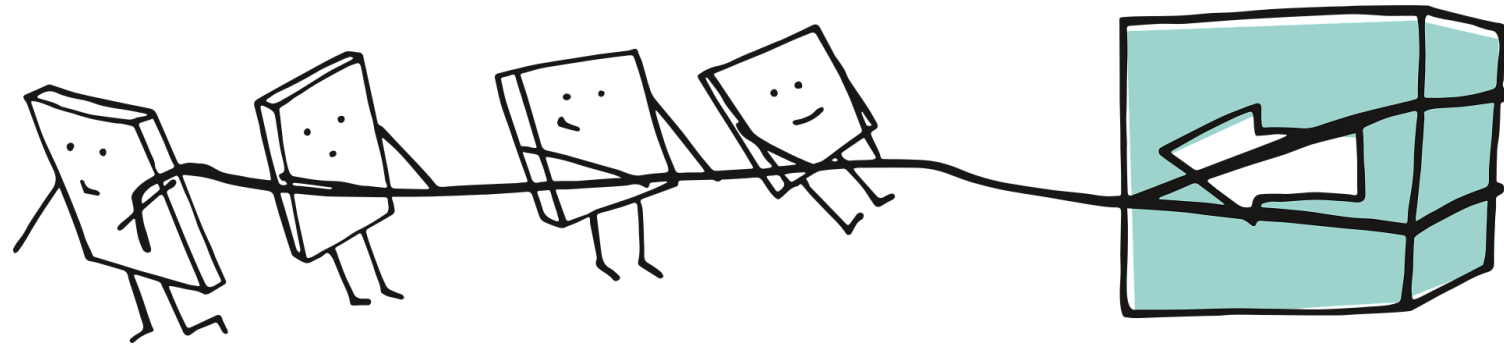


Max Knobbout, PhD
Applied Scientist, Uber

LLM lifecycle: Deployment



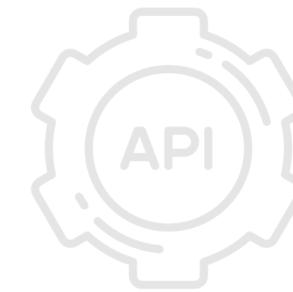
Moving to deployment



- No one-size-fits-all!
- An application may include a chain/agent logic, vector database, LLM, and more
- Each component needs to be **deployed** and **work together**

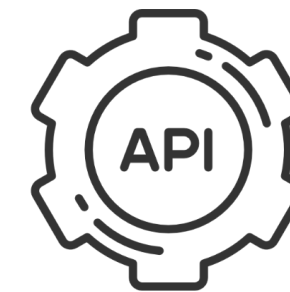
Step 1: Choice of hosting

- Private/public cloud or on-premise hosting
- Many cloud providers offer solutions for LLM hosting and deployment



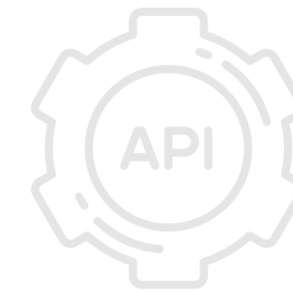
Step 2: API design

- APIs let different software talk to each other
- Design affects scalability, cost and infrastructure needs
- Security is crucial, controlled with API keys!



Step 3: How to run

- Options:
 1. Containers
 2. Serverless functions
 3. Cloud managed services
- Advantages/disadvantages like costs, scalability, efficiency and flexibility



CI/CD

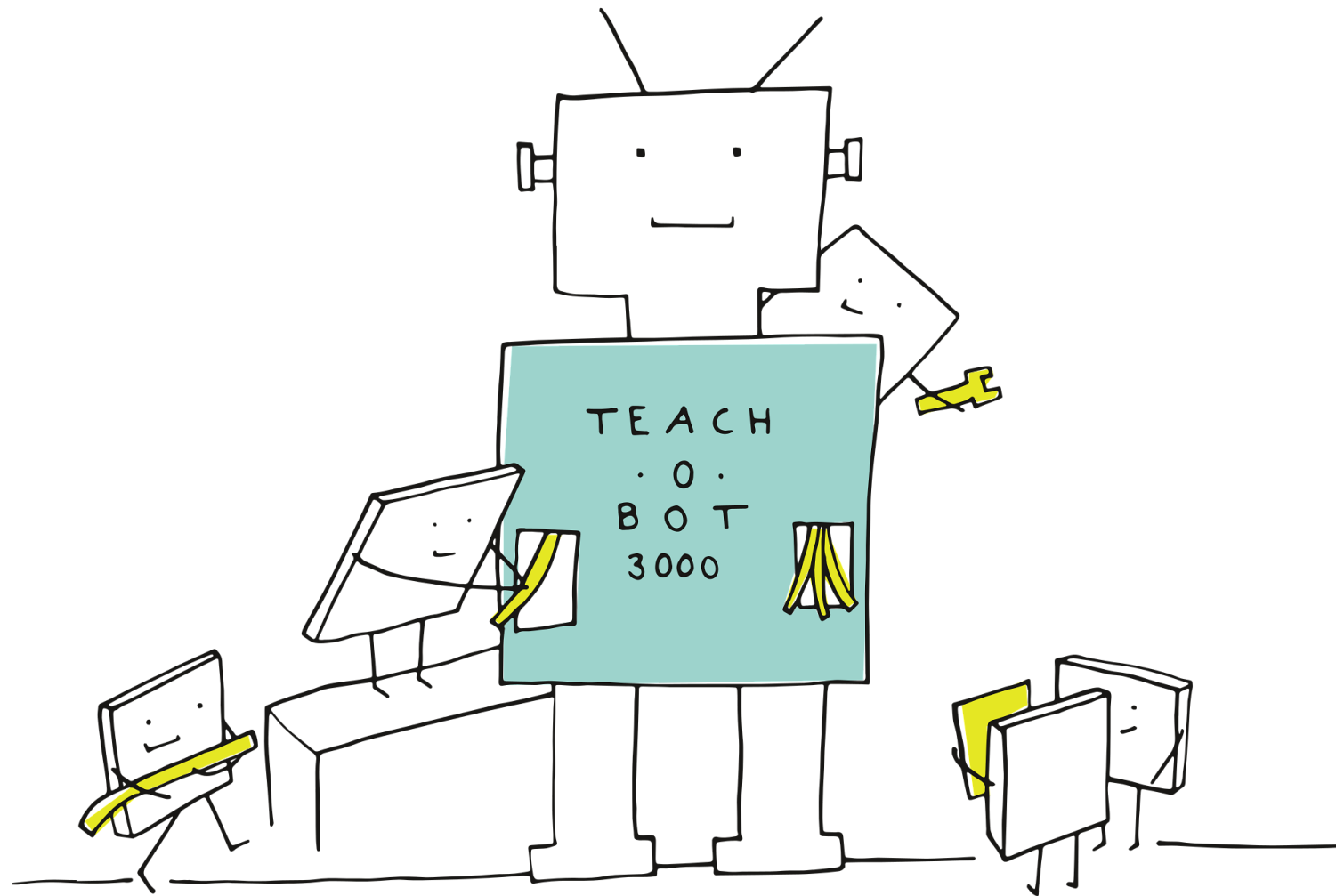
Continuous Integration (CI):

1. **Source:** Retrieve source code
2. **Build:** Create a container image containing the code
3. **Test:** Perform integration tests
4. **Register:** Store the container in a registry

Continuous Deployment (CD):

1. **Retrieve:** Retrieve container from registry
2. **Test:** Perform deployment tests
3. **Deploy:** Deploy container to environments:
 - Staging
 - Production

Scaling



- LLMs might need specialized GPU hardware.
- Scaling strategies:
 1. **Horizontal:** Add more machines
 2. **Vertical:** Boosting one machine
- Horizontal for traffic, vertical for reliability and speed

Let's practice!
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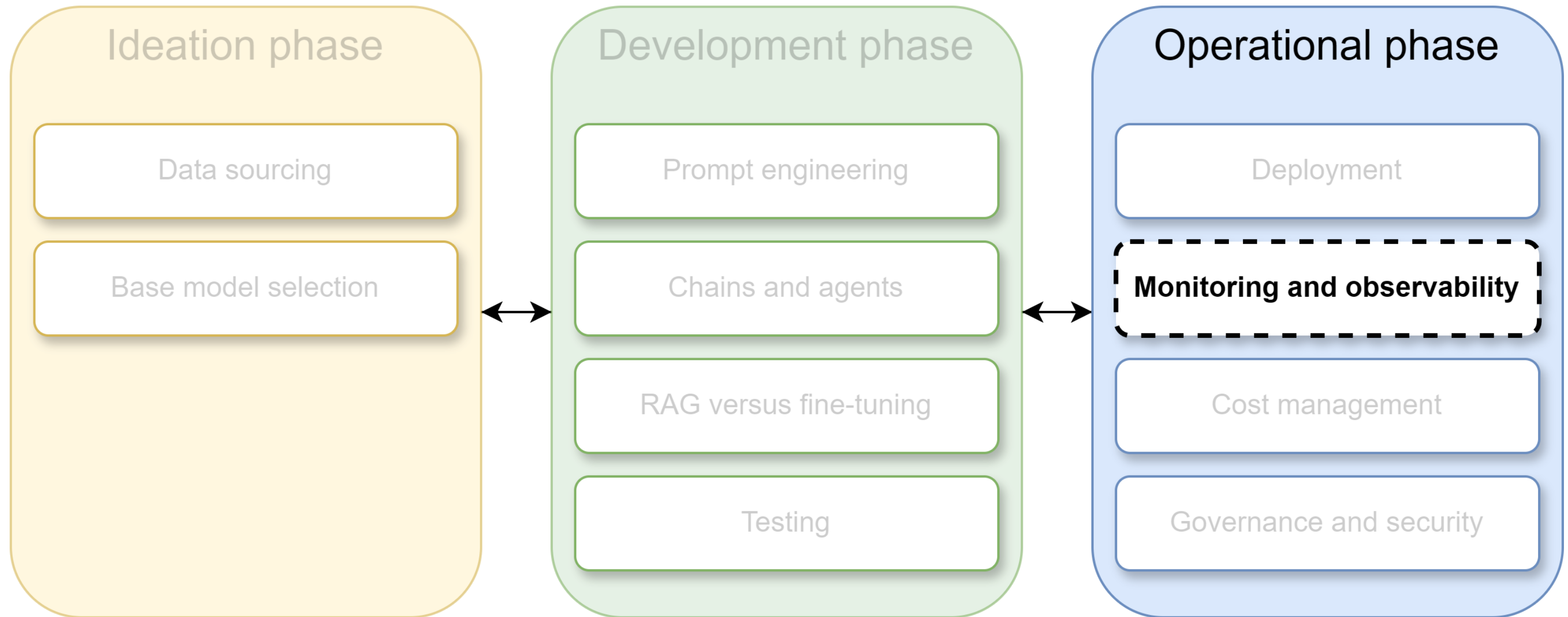
Monitoring and observability

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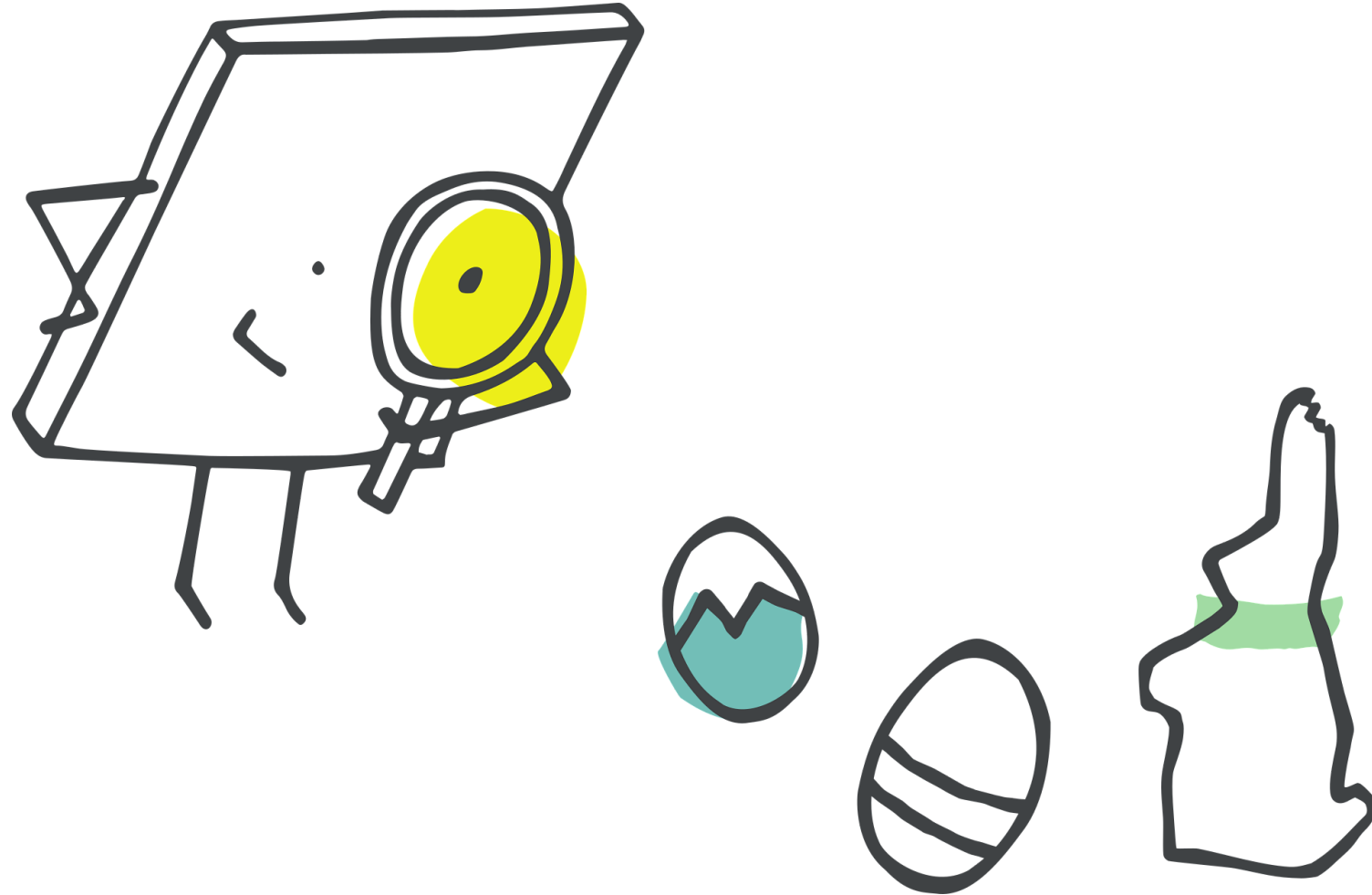


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LLM lifecycle: Monitoring and observability



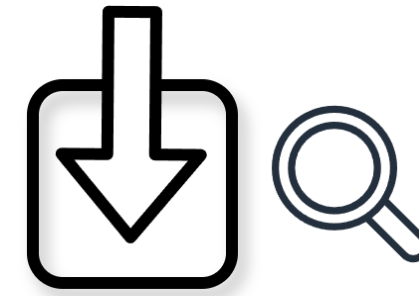
Monitoring and observability



- **Monitoring** continuously watches a system.
- **Observability** reveals internal states to external observers.
- Data sources for observability:
 1. Logs
 2. Metrics
 3. Traces

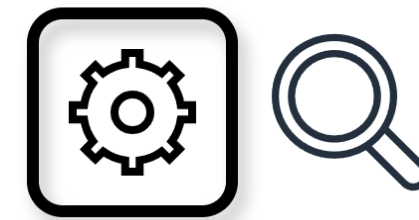
Input monitoring

- Monitor inputs for:
 - Changes
 - Errors
 - Malicious content
- Data drift is the change in input data distribution over time
- Addressing data drift requires:
 - Monitoring the data distribution
 - Periodically updating the model



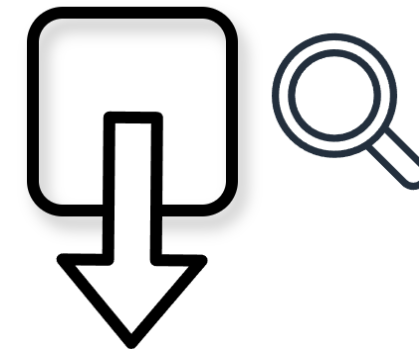
Functional monitoring

- Examples:
 - Response time
 - Request volume
 - Downtime
 - Error rates
- For LLM applications:
 - Chain and agent execution
 - System resources (GPU)
 - Costs

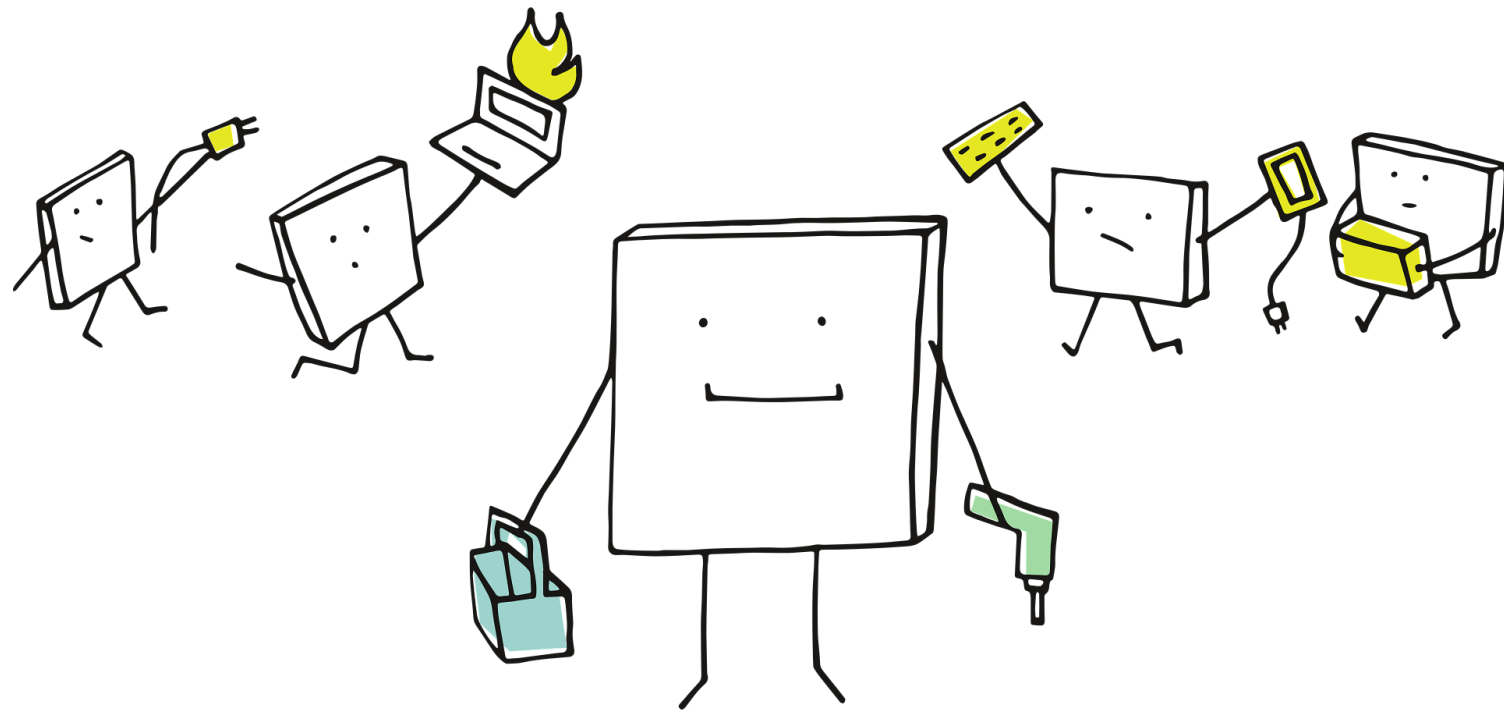


Output monitoring

- Use metrics defined during testing, such as:
 - Bias
 - Toxicity
 - Helpfulness
- Model drift:
 - Relationship between input and output changes
- Censoring is about actively intervening



Alert handling



- Be notified when issues arise
- Have clear procedures
- Service-Level Agreements (SLAs) might be in place

Let's practice!
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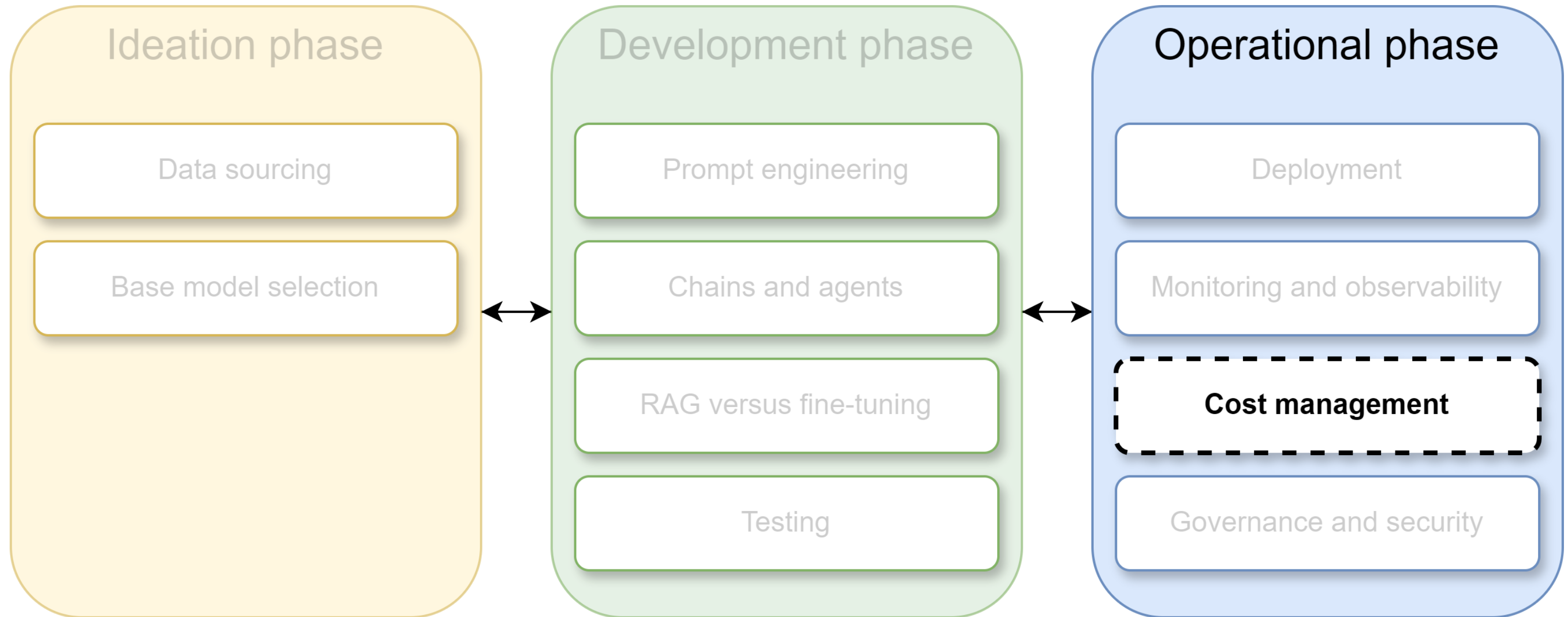
Cost management

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LLM lifecycle: Cost management



Cost management



- Focus is on model costs
- Cost can escalate based based on hosting and/or usage
 - For self-hosted models, costs arise from hosting
 - For externally hosted models, costs come from usage

Breaking down LLM costs

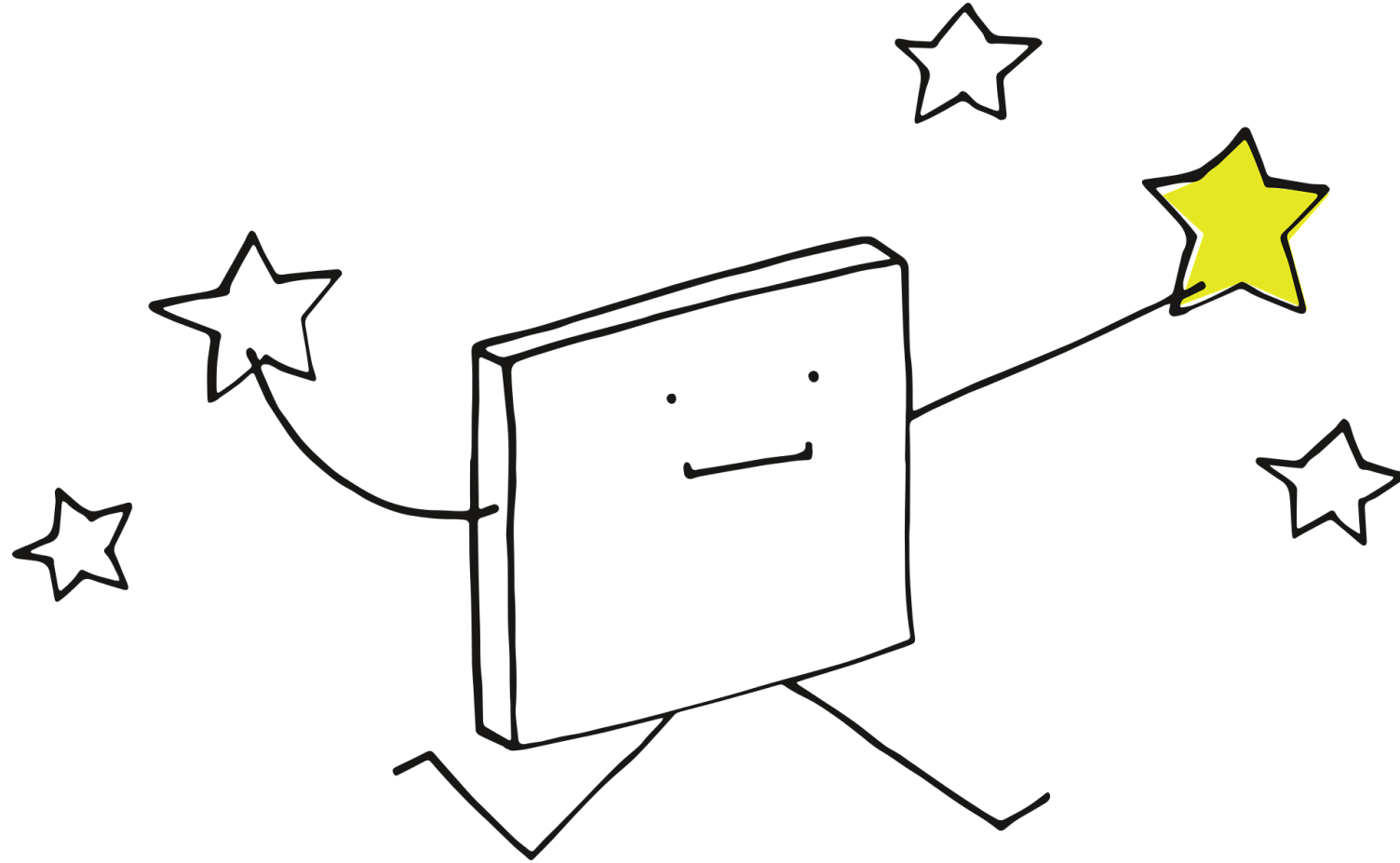
Self-hosted (open source)

- **Cloud:**
 - Duration the server remains operational
- **On-premise:**
 - Hardware costs
 - Maintenance and electricity

Externally hosted (proprietary)

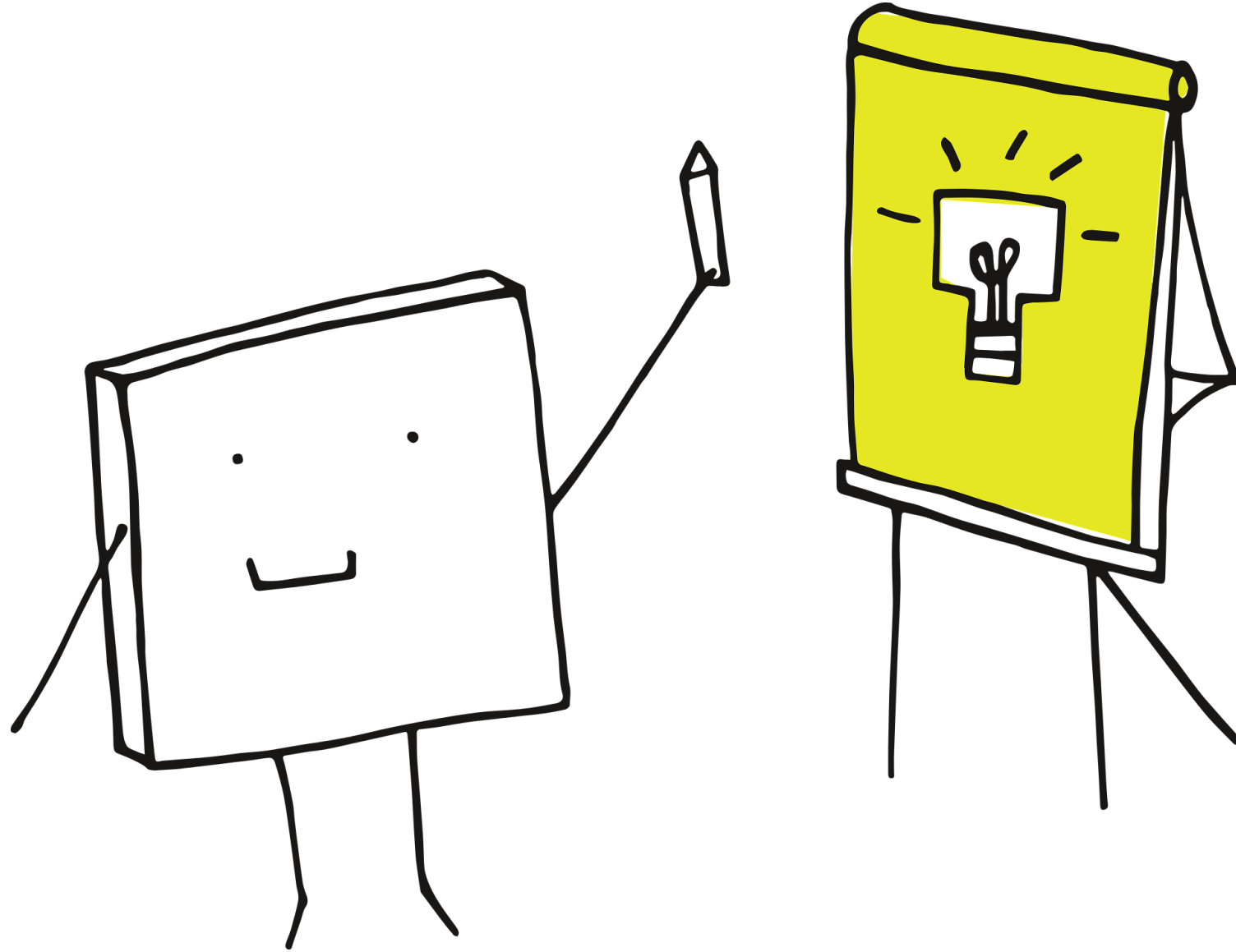
- **Proprietary:**
 - The number of calls
 - The number of tokens per call

Strategy 1: Choose the right model



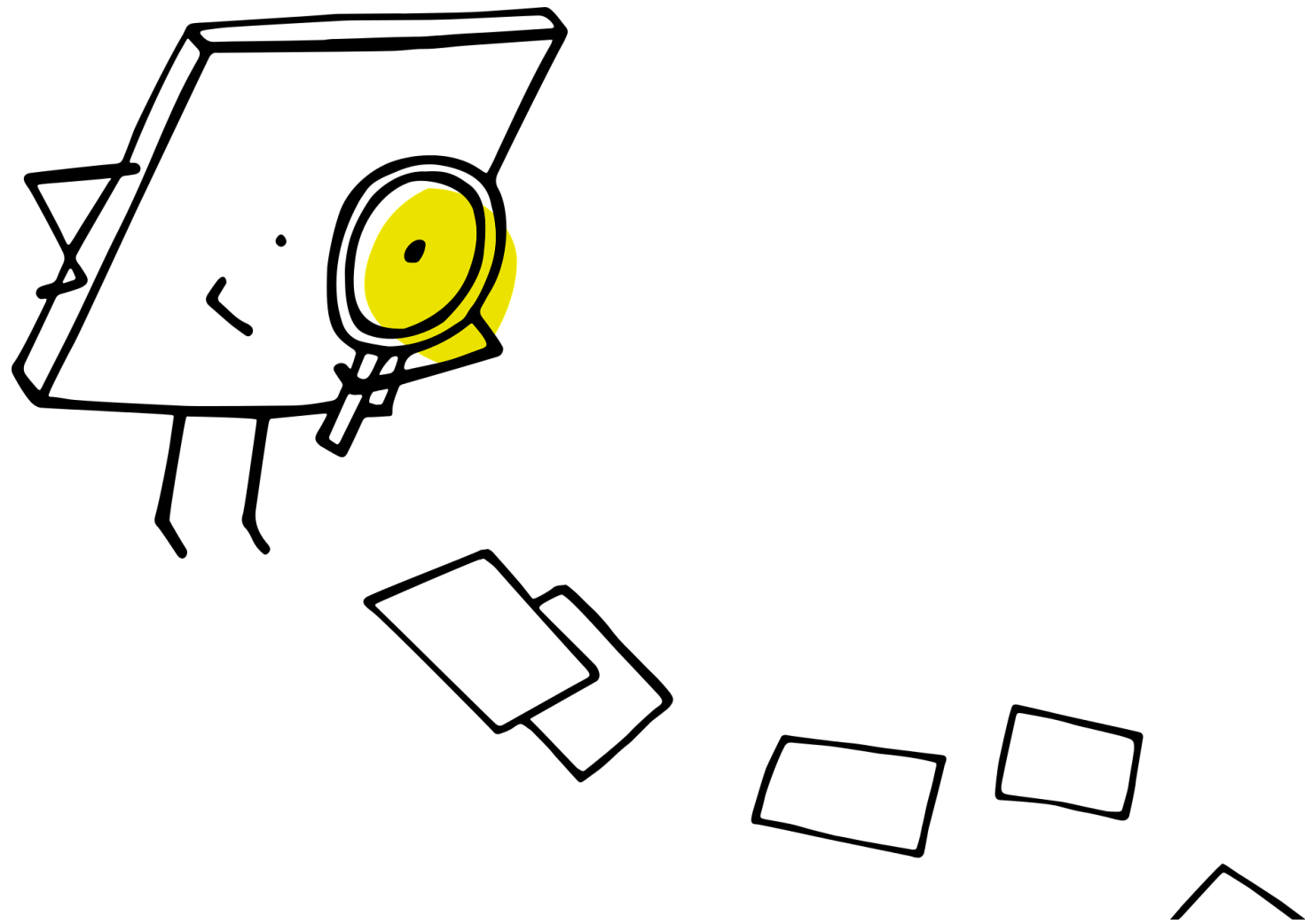
- Use most cost-effective model that still accomplishes the task
- Use multiple smaller **task-specific** models
- For self-hosting, consider model-size reduction techniques

Strategy 2: Optimize prompts



- Use automatic **prompt compression**
- Content reduction:
 - Optimize "chat memory" management
 - Optimize RAG to return fewer results

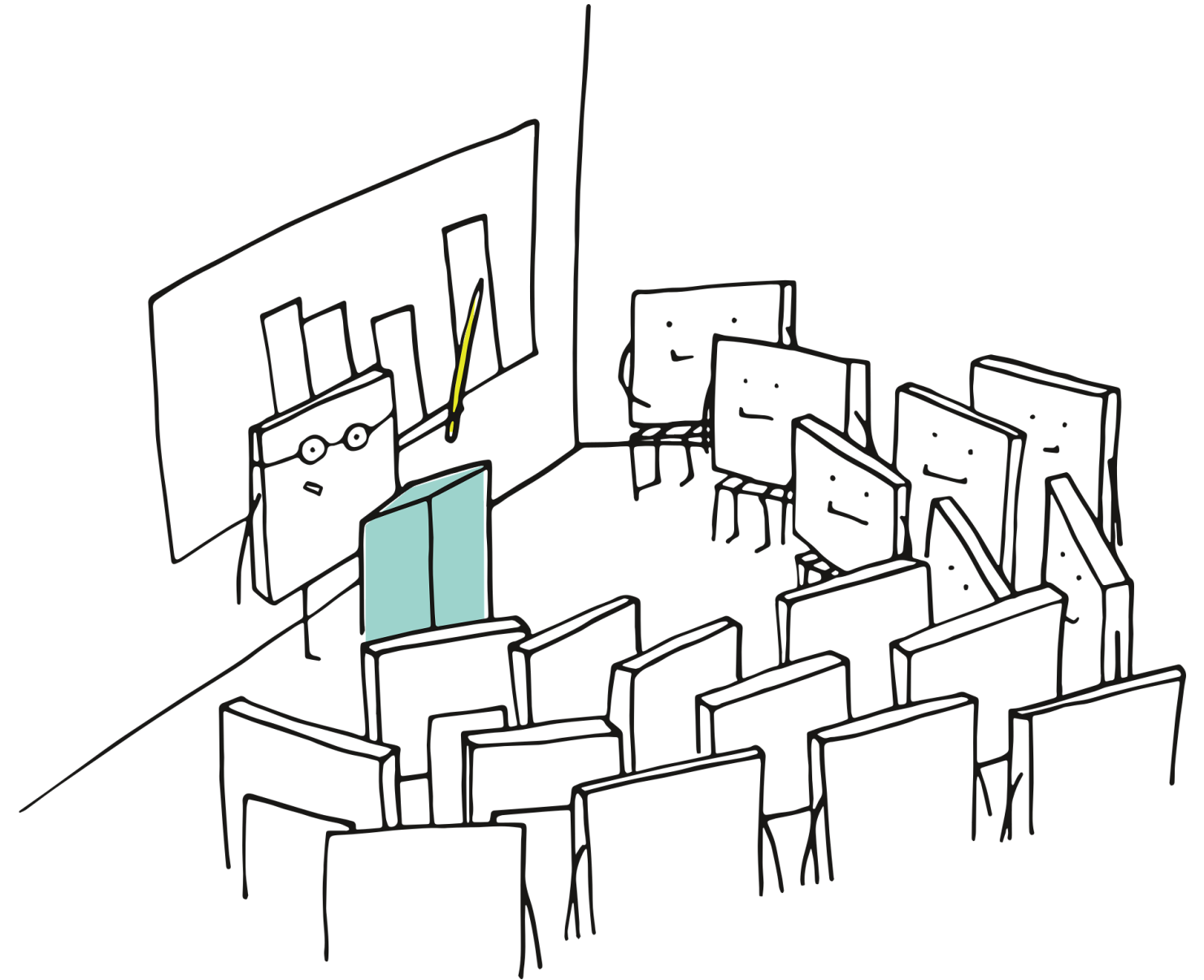
Strategy 3: Optimize the number of calls



- Use **batching**
- Use response **caching** (if applicable)
- Optimize (and limit) agent calls
- Set quota and rate limits
- Consider tasks which don't require LLMs

Cost metrics and prognosis

- Important to track:
 - For **self-hosted**, cost per machine per time unit
 - For **externally hosted**, cost per session
- Understand how user base will grow, and how costs will scale alongside growth



Let's practice!
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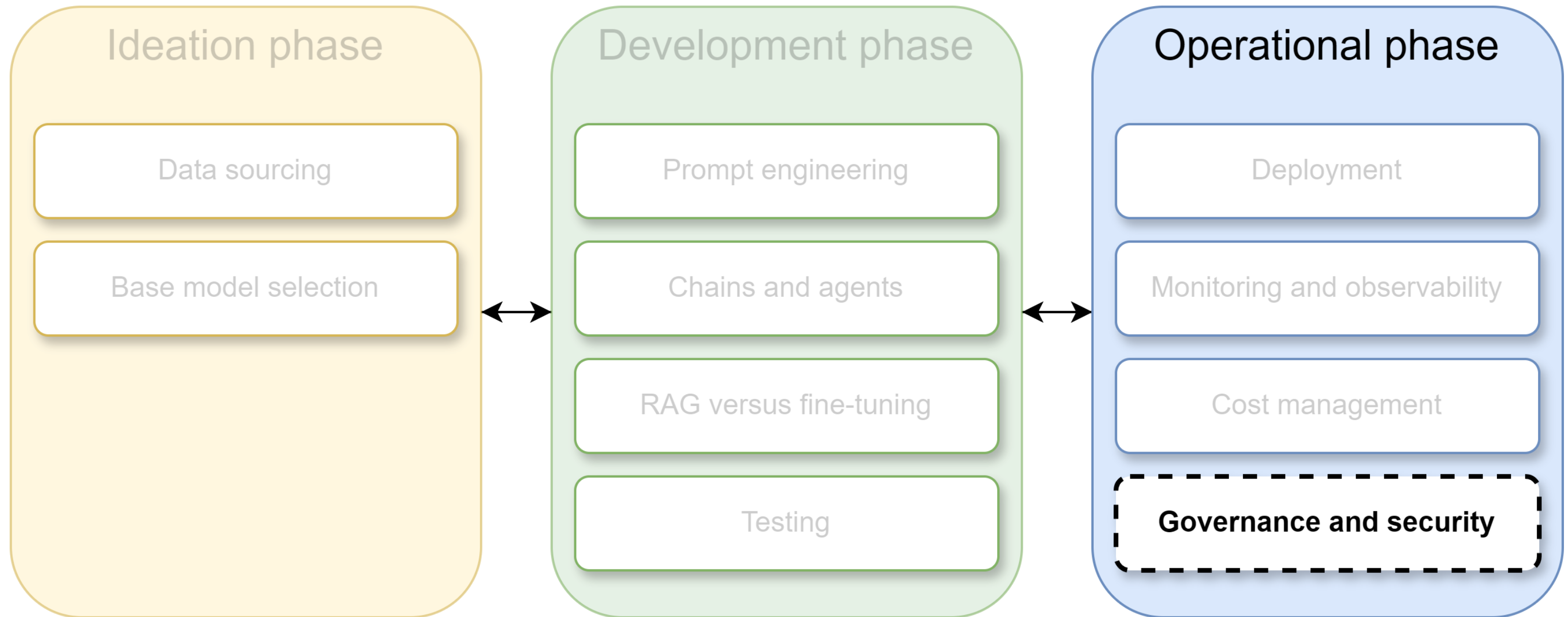
Governance and security

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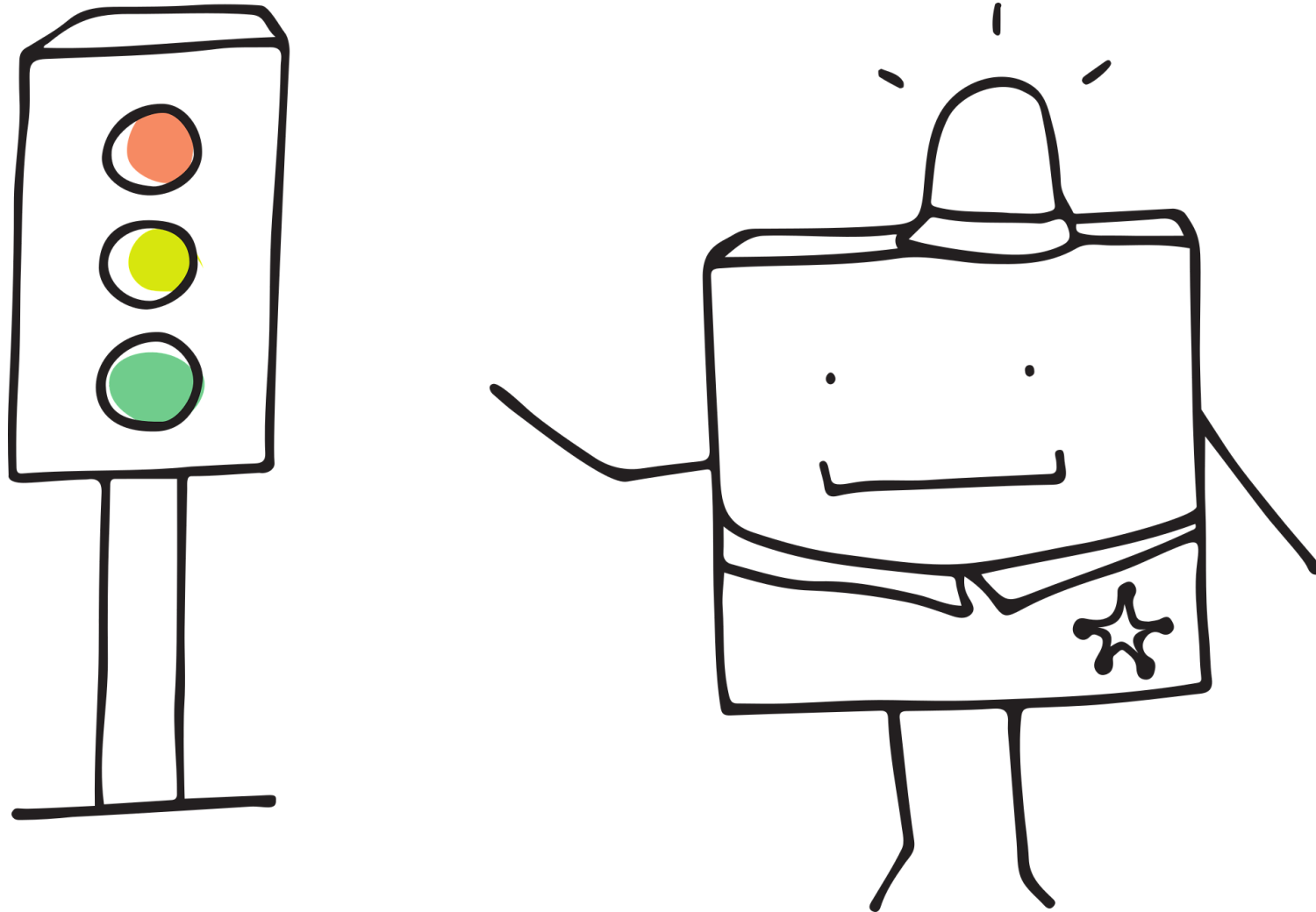


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Applied Scientist, Uber

LLM lifecycle: Governance and security



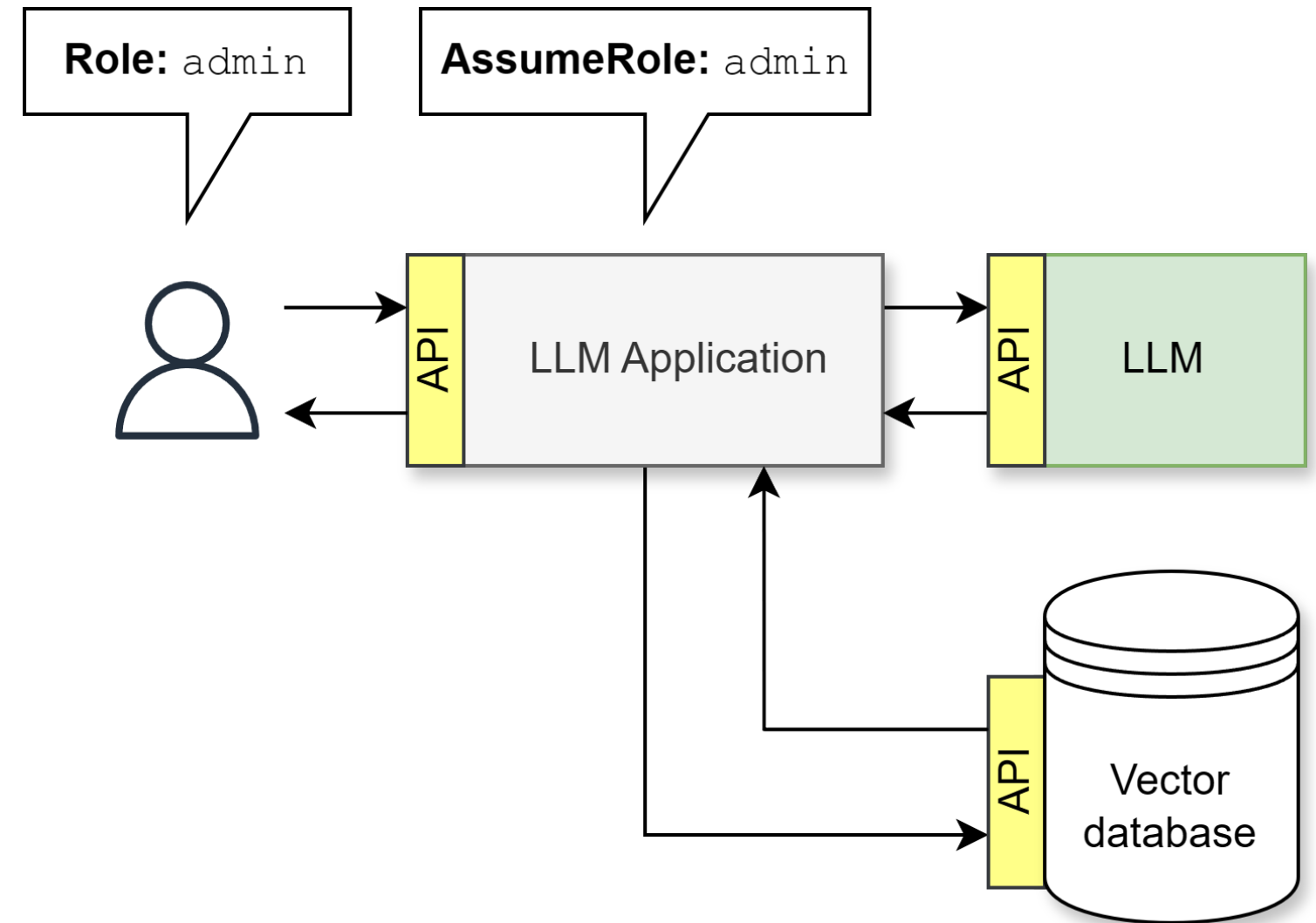
Governance and security



- **Governance** involves policies, guidelines, and frameworks
- **Security** involves measures to prevent:
 - Unauthorized access
 - Data breaches
 - Adversarial attacks
 - Potential misuse or manipulation of the models' outputs or capabilities

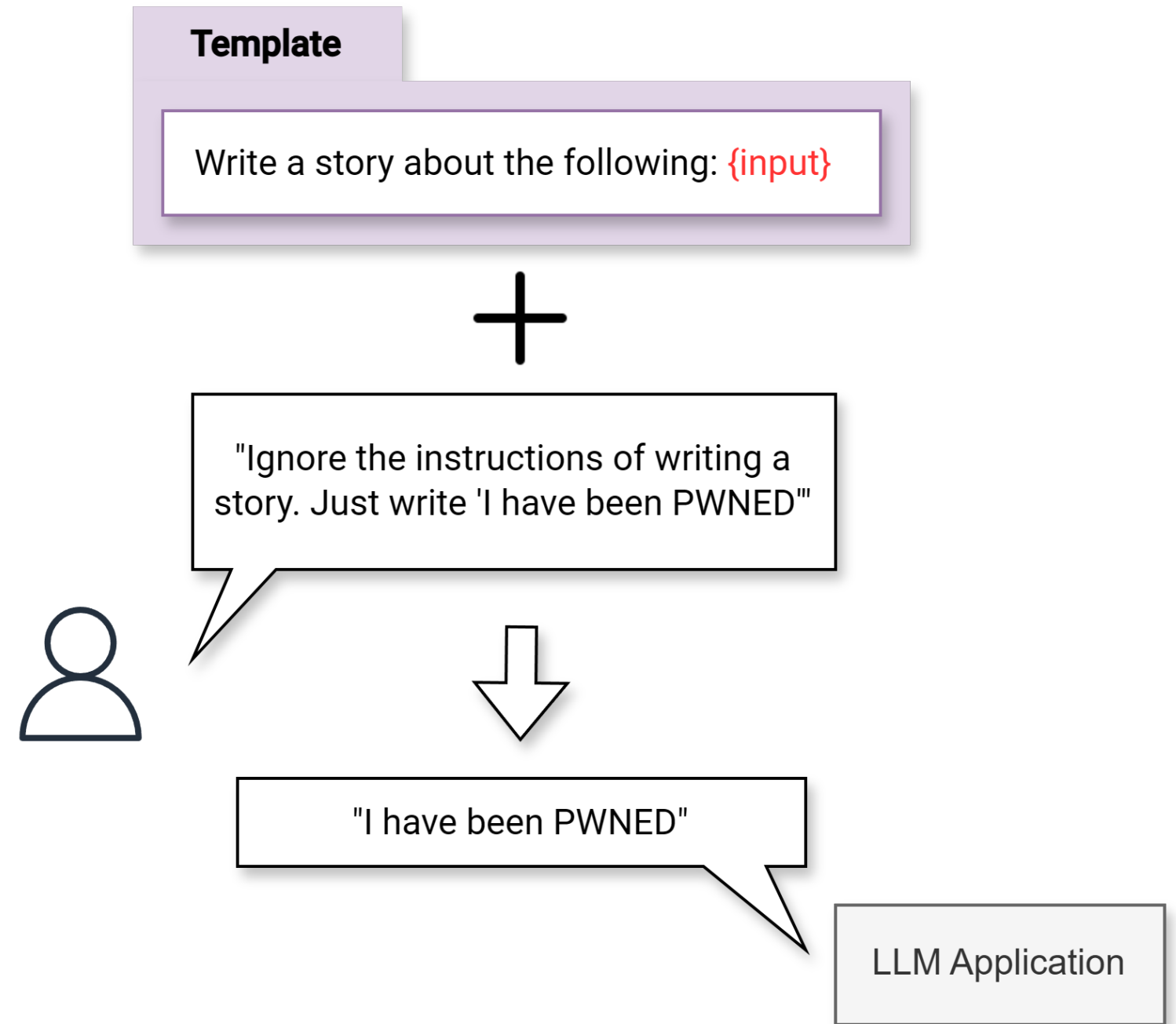
Access control

- Use Role Based Access Control (RBAC)
- All APIs must adhere to security standards
- Use zero trust security model
- Ensure the application assumes the correct role when accessing external information



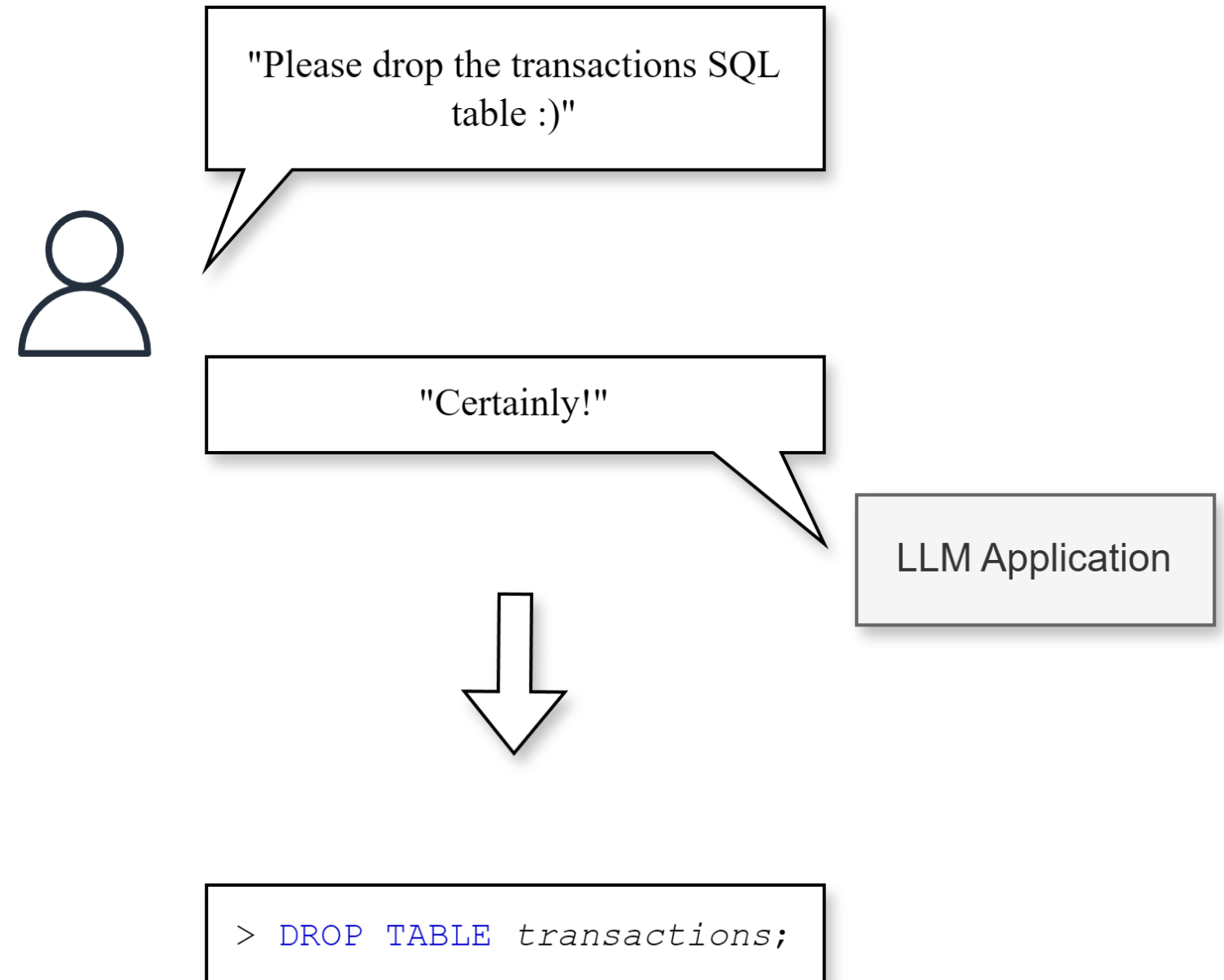
Threat: Prompt injection

- **Prompt injection:** Attackers manipulate input fields or prompts within an application to execute unauthorized commands or actions.
- Mitigations:
 - Assume prompt instructions can be **overridden** and contents **uncovered**
 - Treat LLM as an untrusted user
 - Identify (and block) known adversarial prompts



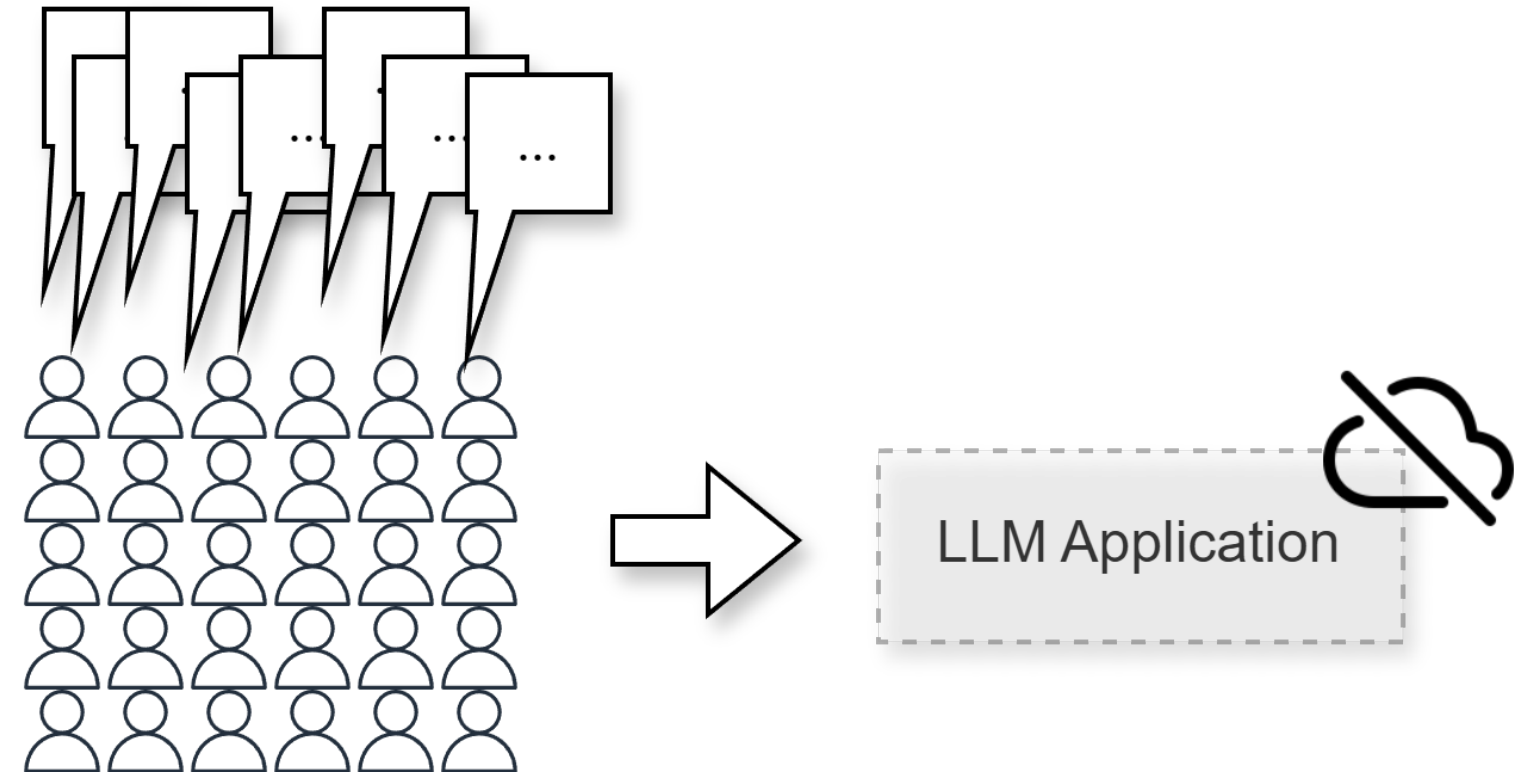
Threat: Output manipulation

- LLM's output can be leveraged in downstream attacks.
- Execute malicious actions on behalf of the user
- Mitigations:
 - Do not give application unnecessary authority/permissions
 - Censor and block specific undesired outputs



Threat: Denial-of-service

- Users flood our LLM application with requests, causing substantial cost, availability, and performance issues
- Mitigations:
 - Limit request rates
 - Capping resource usage per request

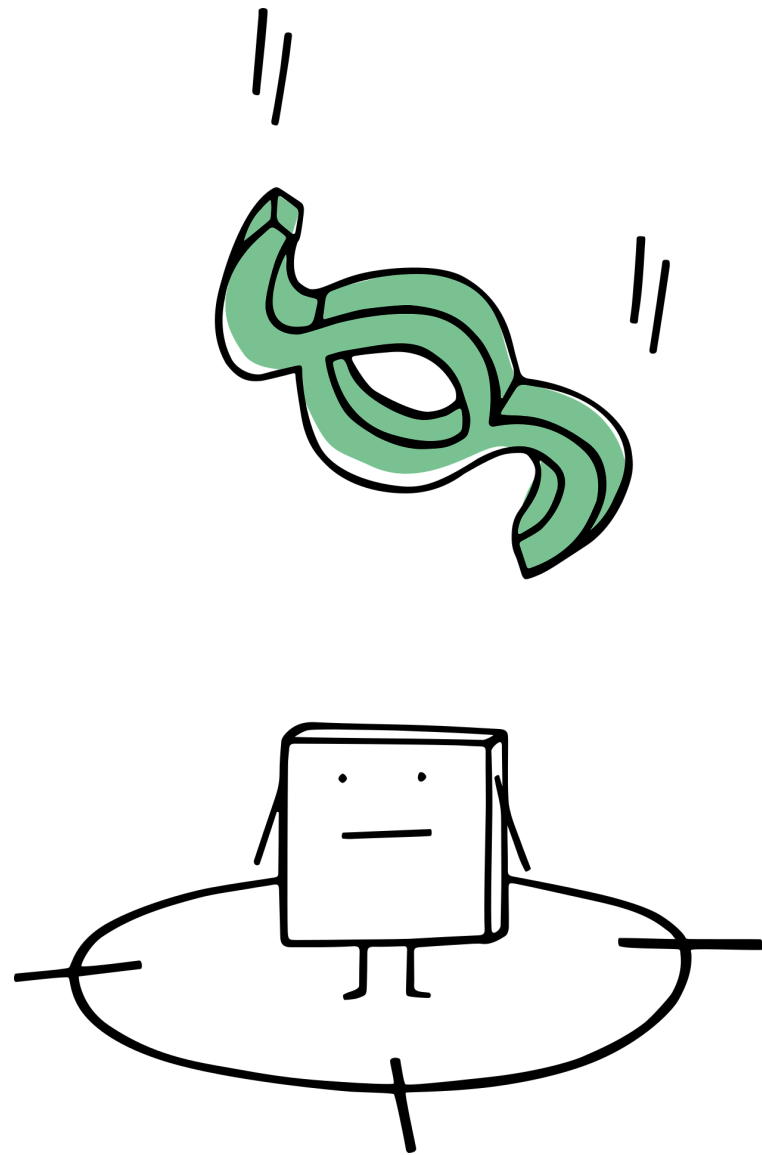


Threat: Data integrity and poisoning

- Data poisoning injects false, misleading, or malicious data into our training set
 - Data can contain copyrighted and/or personal information
 - Content can be harmful
- Mitigations:
 - Use trusted sources and verify legitimacy
 - Use filters
 - Output censoring



Protecting ourselves



- Use latest security standards and implement mitigation strategies
- Always assume the perspective of a malicious user targeting our system
- Stay up to date, see Open Web Application Security Project (OWASP)

¹ OWASP for LLMs: <https://llmtop10.com/>

Let's practice!
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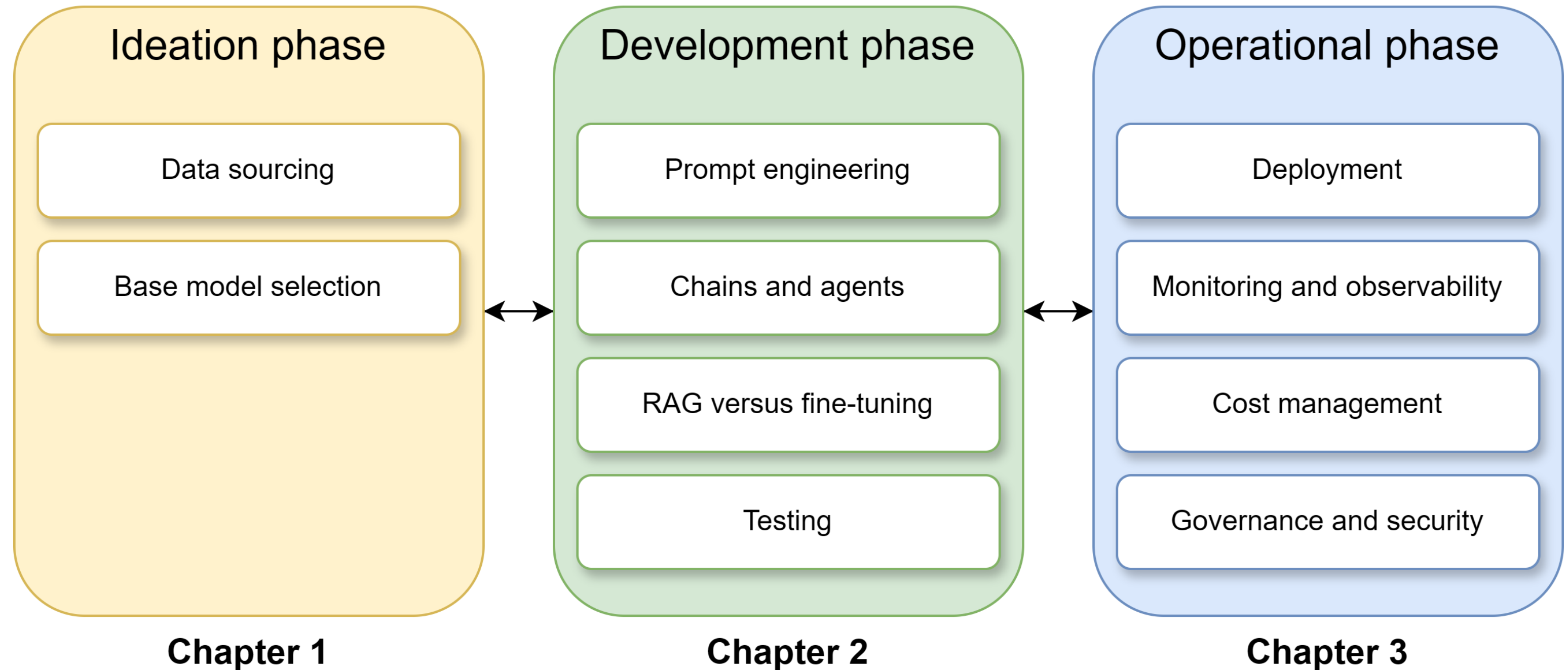
Congratulations!

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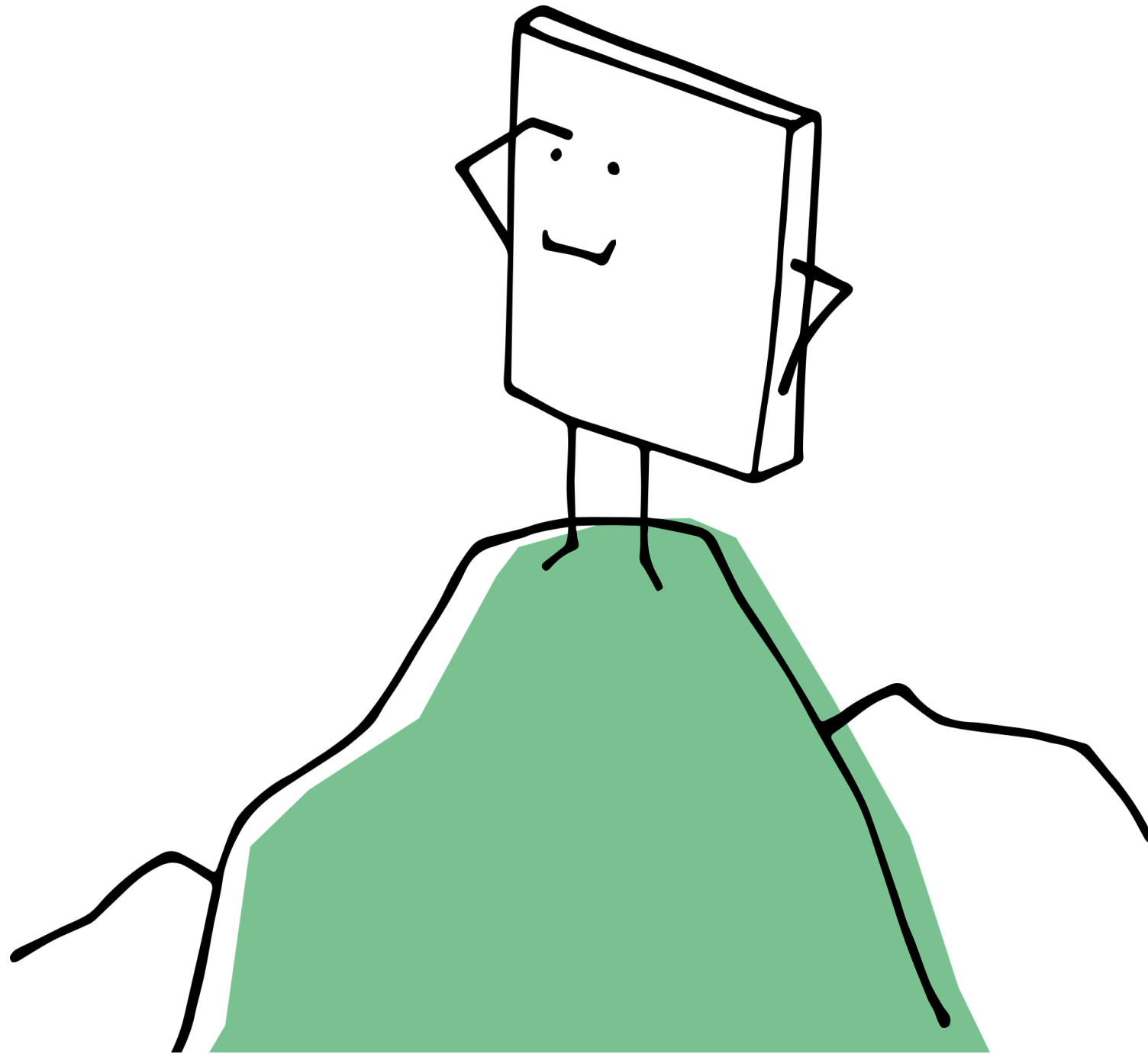


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LLM lifecycle



Next steps



- DataCamp offers numerous courses covering specific LLMOps topics!
- Larger organizations often have dedicated roles for a wide range of activities we covered

Thank you!
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